

Open Data Quality

Assessment and Evolution of (Meta-)Data Quality
in the Open Data Landscape

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Contents

- **Preliminaries:** Open Data Landscape and Portals
- **Problem** Statement and **Motivation**
- Quality **Metrics**
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What is Open Data?

Freely available data,

open access, preferable on the WWW

published in an **open** and **machine readable format**

e.g., CSV, JSON, RDF

which allows **everybody**

open license which allows use, reuse, modification, redistribution

to do **everything without restrictions**

private, non-commercial and commercial

at anytime

24/7

The Open Data Landscape

Cities, International Organizations, National and European Portals:



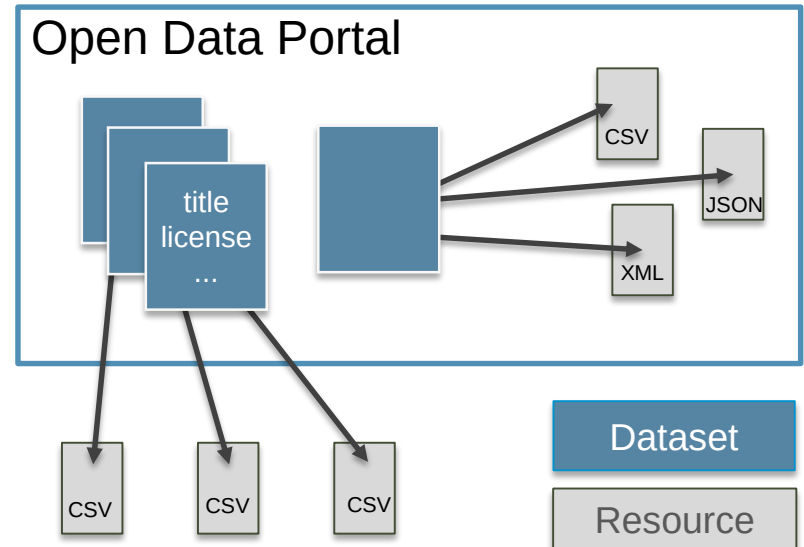
Open Data Portals

Single point of access

Meta data

- Licenses
- Provenance
- Formats
- ...

Typical software



E.g.: data.gv.at

Open Data Portal by the
Austrian Government

```
https://www.data.gv.at/katalog/api/rest/dataset

[
  "abaenderungsantrag-nr",
  "abfallentsorgung-containerstandpl-tze-engerwitzdorf",
  "abfallentsorgung-engerwitzdorf",
  "abfluss-und-seepegel-land-salzburg",
  "abwasser-behandlungsanlagen-land-salzburg",
  "abwasserwirtschaft-engerwitzdorf",
  "adressen-tirol",
  "adressinformationen-tirol",
  "adressliste-gemeinnutziger-bauvereinigungen",
  "aerzte-engerwitzdorf",
  "agrarstrukturhebung-2010-uberblick",
  "aktuelleozondatenoesterreich",
  "aktuelle-verkehrsbehinderungen",
  "alkoholberatungsstellen-in-oberosterreich",
  "allg-bildende-pflichtschulen-in-oo",
  "allgemein-bildende-hohe-schulen",
  "allgemein-bildende-hohe-schulen-in-tirol",
  "allgemein-bildende-hohe-schulen-in-oberosterreich",
  "allgemein-bildende-hohe-schulen-in-oberosterreich-bewil",
  "allgemein-bildende-hohe-schulen-in-oberosterreich-strassen",
]
```

Titel	Veröffentlichende Stelle / Datenverantwortliche Stelle	Veröffentlicht auf data.gv.at am	Letzte Änderung	Format
Bevölkerung nach Alter und Geschlecht Bevölkerung nach Alter, Geschlecht und Wohngemeinde zum 1.1. des jeweiligen Jahres	Land Niederösterreich / Abteilung Raumordnung und Regionalpolitik- Statistik	11.04.2013	10.05.2015	CSV
Bevölkerung nach Gemeinden - Volkszählungen Bevölkerung nach Wohngemeinden zum Volkszählungstichtag des jeweiligen Jahres	Land Niederösterreich / Abteilung Raumordnung und Regionalpolitik- Statistik	11.04.2013	10.05.2015	CSV
Landtagswahl 2013 Wahlergebnisse der Landtagswahl 2013 \ der Hinweis in der Attributbeschreibung ! N a c h f o l g e ...	Land Niederösterreich / Abteilung Staatsbürgerschaft und Wahlen	28.10.2014	10.05.2015	CSV
Wanderungen nach Gemeinden Zuzüge, Wegzüge und Gemeindebinnenwanderungen im jeweiligen Jahr	Land Niederösterreich / Abteilung Raumordnung und Regionalpolitik- Statistik	11.04.2013	10.05.2015	CSV

CKAN Metadata (JSON)

```
d: {  
  "license_title": "Creative Commons Namensnennung", "maintainer": "Stadtvermessung Graz",  
  "author": "",  
  "author_email": "stadtvermessung@stadt.graz.at",  
  "resources": [  
    {  
      "size": "6698",  
      "format": "CSV",  
      "mimetype": "",  
      "url": "http://data.graz.gv.at/.../Bibliothek.csv"  
    }  
  ], "tags": [  
    "bibliothek",  
    "geodaten",  
    "graz",  
    "kultur",  
    "poi" ],  
  "license_id": "CC-BY-3.0",  
  "organization": null,  
  "name": "bibliotheken",  
  "notes": "Standorte der städtischen Bibliotheken...",  
  "extras": {  
    "Sprache des Metadatensatzes": "ger/deu Deutsch"  
  },  
  "license_url": "http://creativecommons.org/.../by/3.0/at",  
}
```

core keys

resource keys

extra keys

What is the Problem?

There is a concern of **quality issues** on data portals [1]:

Metadata

- Missing values
- Incorrect values
- No contact info
- Wrong/missing file format description

Resources

- Changing URLs
- Formats (e.g. CSV not RFC 4180 compliant -> [,;\t#])
- Encoding (e.g., mixed)

Hypothesis

Objective Quality Metrics

- **discover**, point out and **measure** quality and heterogeneity issues in data portals

Automated Quality Assessment Framework

- **monitor** and assess the **evolution** of quality metrics over time

Quality Metrics

Metrics

Dimensions	Description
Retrievability	The extent to which meta data and resources can be retrieved.
Usage	The extent to which available meta data keys are used to describe a dataset.
Completeness	The extent to which the used meta data keys are non empty.
Accuracy	The extent to which certain meta data values accurately describe the resources.
Openness	The extent to which licenses and file formats conform to the open definition.
Contactability	The extent to which the data publisher provide contact information.

Objective measures which can be **automatically computed** in a **scalable** way

Concrete Metrics (1/2)

Retrievability:

- HTTP GET lookup for datasets (API) and resources

Usage:

- Ratio of used keys and all identified keys (on a data portal)

Completeness:

- Ratio of non-empty keys in a dataset

Concrete Metrics (2/2)

Openness:

- Licenses: map to list by opendefinition.org
- Formats: pre-defined set of file formats, e.g. CSV, XML, ...

Contactability:

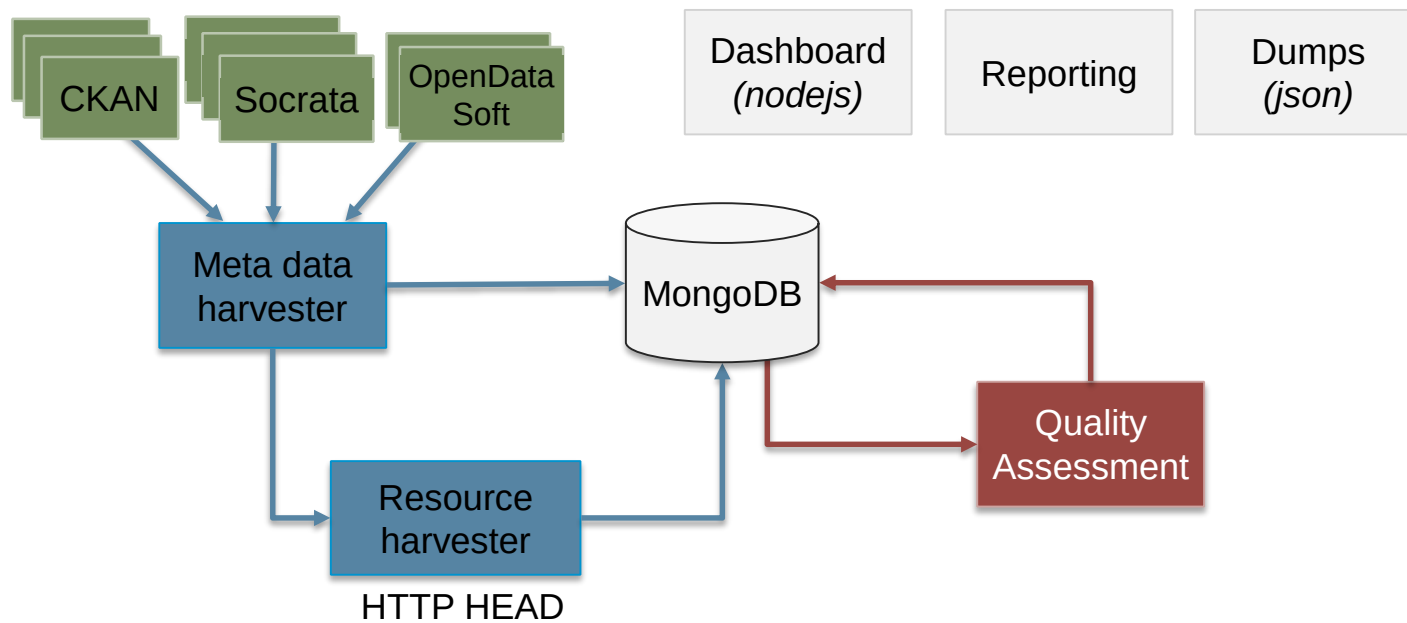
- Availability of contact information: (i) text, (ii) url, (iii) email

Accuracy:

- Formats, file size, mime-type
- Currently based on respective HTTP response header fields

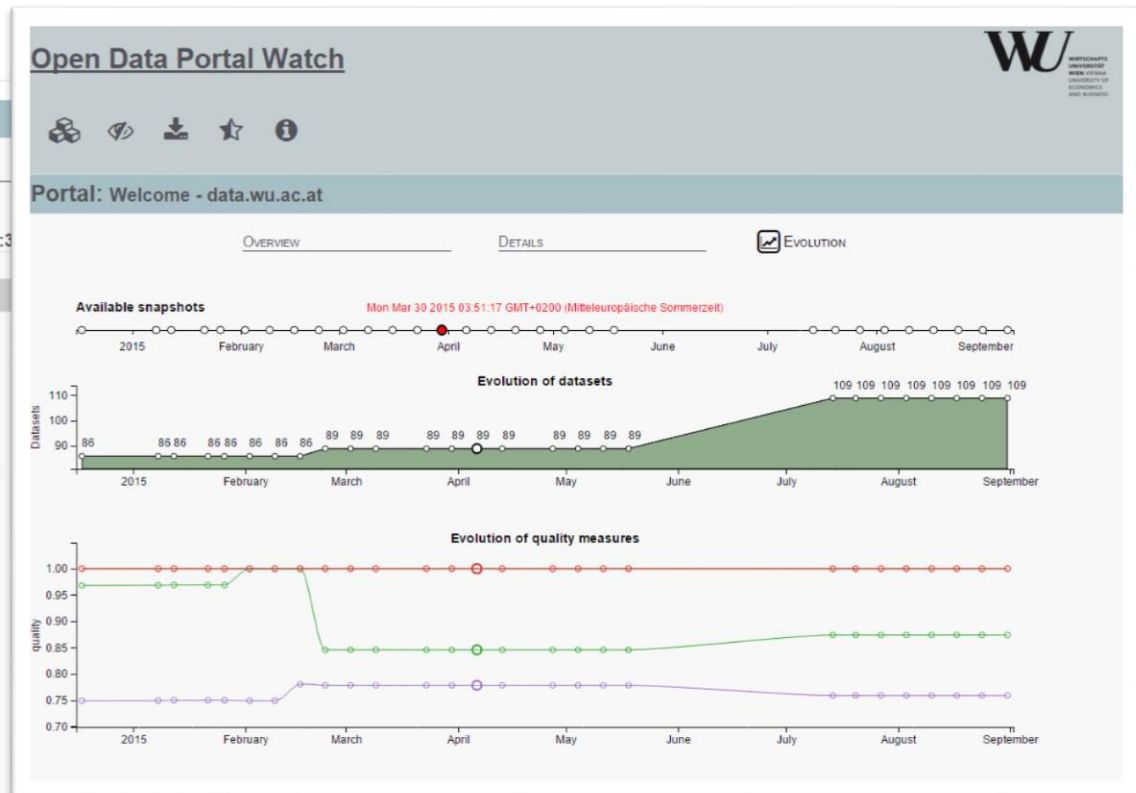
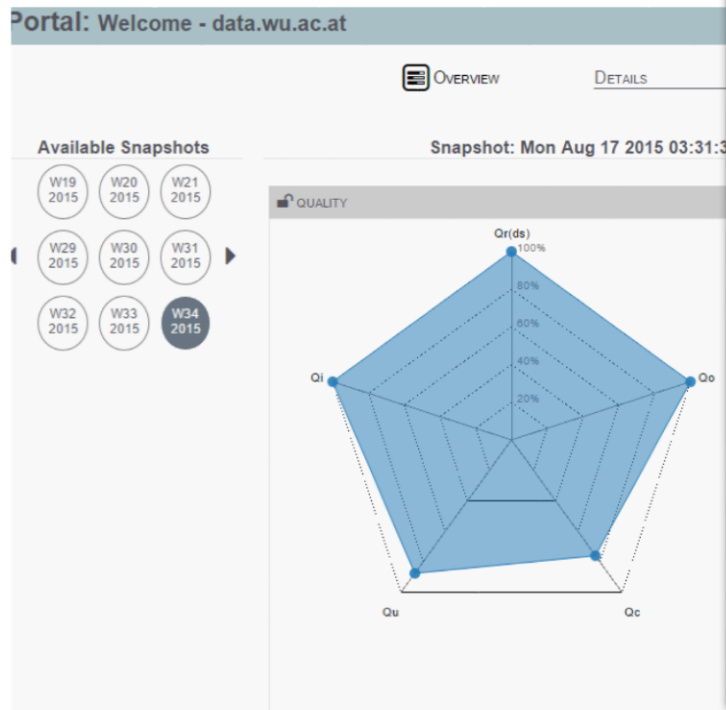
Automated QA Framework

Architecture



Open Data Portal Watch

Scalable quality assessment & monitoring framework for Open Data Portals



Findings

Portals Overview

Based on 126 CKAN data portals:

$ \mathcal{D} $	$ \mathcal{R} $	$ \mathcal{K}^c $	$ \mathcal{K}^e $	$ \mathcal{K}^r $	Licenses	Formats	Tags
745196	3120119	240	3634	366	218	1516	173586

Top 5 (wrt. datasets):

URL	Origin	Software	$ \mathcal{D} $	$ \mathcal{R} $	C.-L. ^[a]	% ^[b]	C.-L. ^[c]
www.data.gc.ca	Canada	CKAN	244602	1164896	522283	44.8	7814 GB
data.gov	US	CKAN	160049	569180	129003	22.7	967 GB
data.noaa.gov	US	CKAN	64359	619544	127524	20.6	1664 GB
geothermaldata.org	US	CKAN	56179	62406	26206	42	163 GB
publicdata.eu	Europe	CKAN	55459	140278	69325	49.4	189 GB

3.12M URL values, **1.92M** distinct, **1.91M** are syntactically valid URLs

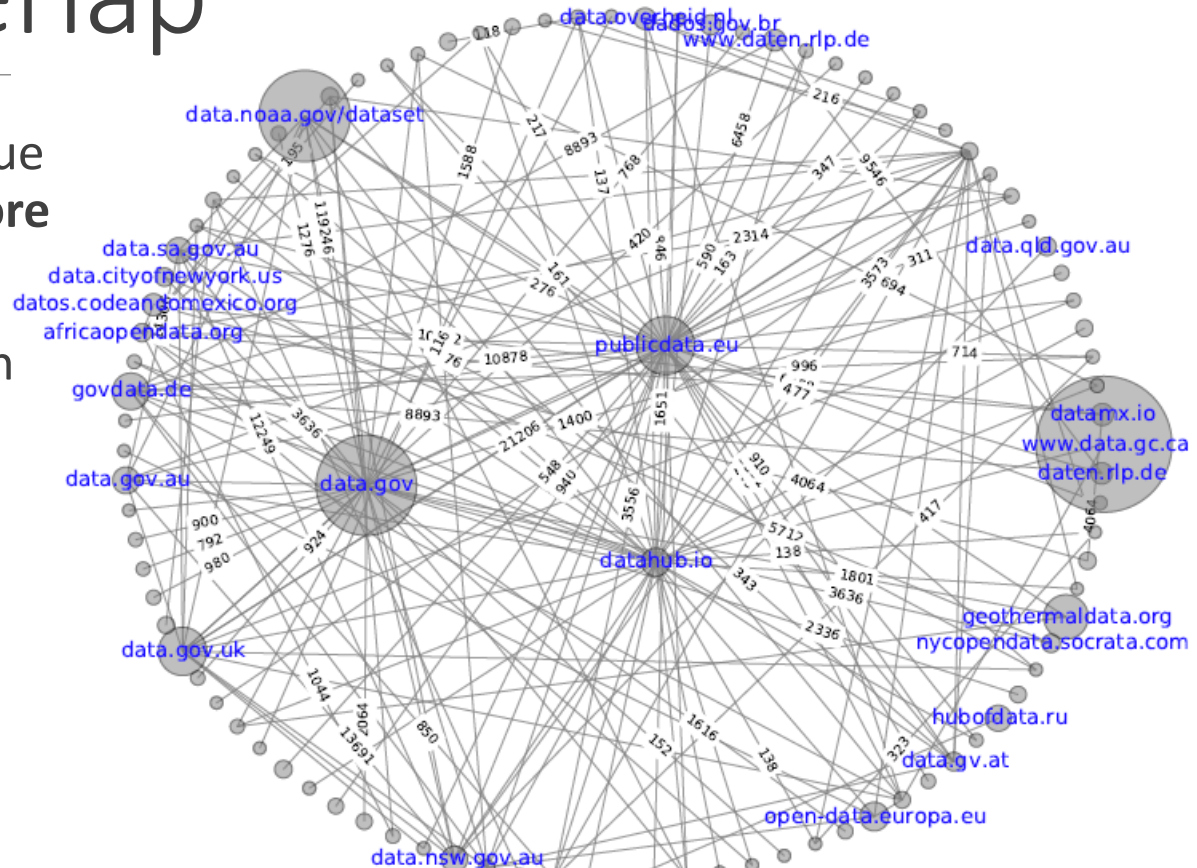
1.1M Content-Length HTTP header fields resulting in **12.297 TB**

Portal Overlap

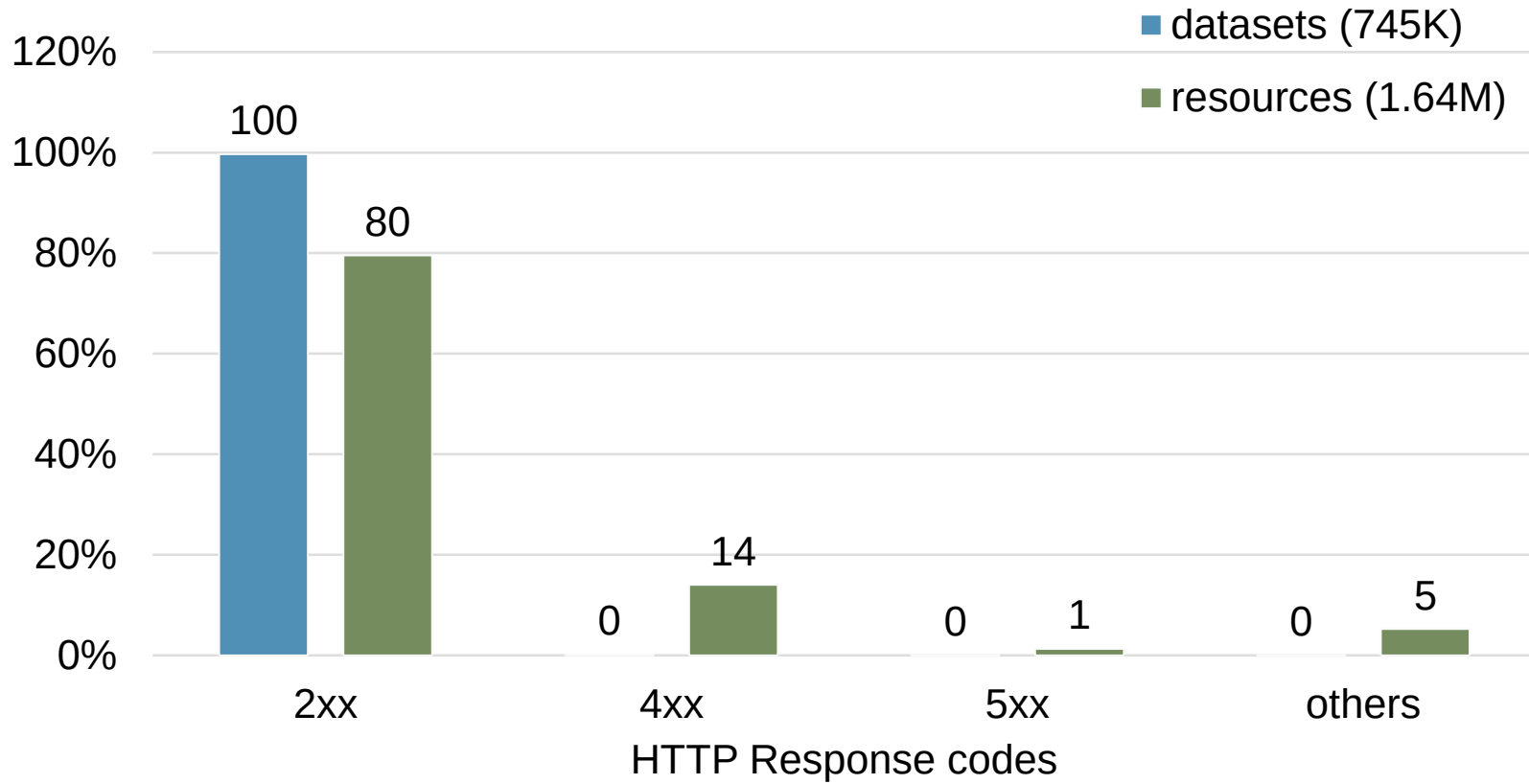
13% (260K) of the unique resources appear in **more than one dataset**

12% (227K) resources in **more than one portal**

biggest portals act as parent/**harvester** portals (e.g. data.gov, publicdata.eu)



Retrievability



Openness

Top 10 licenses and formats over all portals:

license_id	Nº	%	p	format	Nº	%	p
ca-ogl-lgo	244597	48.6	1	PDF	679440	25.6	89
notspecified	55918	11.1	65	HTML	611313	23	70
cc-by	51800	10.3	76	other	182963	6.9	6
uk-ogl	25585	5.1	16	CSV	153617	5.8	107
us-pd	21806	4.3	1	originator data format	115434	4.3	1
empty	11858	2.4	23	geotif	95323	3.6	2
dl-de-by-1.0	8862	1.8	4	XML	84571	3.2	79
cc0	7254	1.4	1	ZIP	69696	2.6	76
cc-nc	6393	1.3	28	tiff	66045	2.5	10
dl-de-by-2.0	6180	1.2	2	SHAPE	57565	2.2	12
others	63066	12.5		others	542969	20.4	

confirmed open

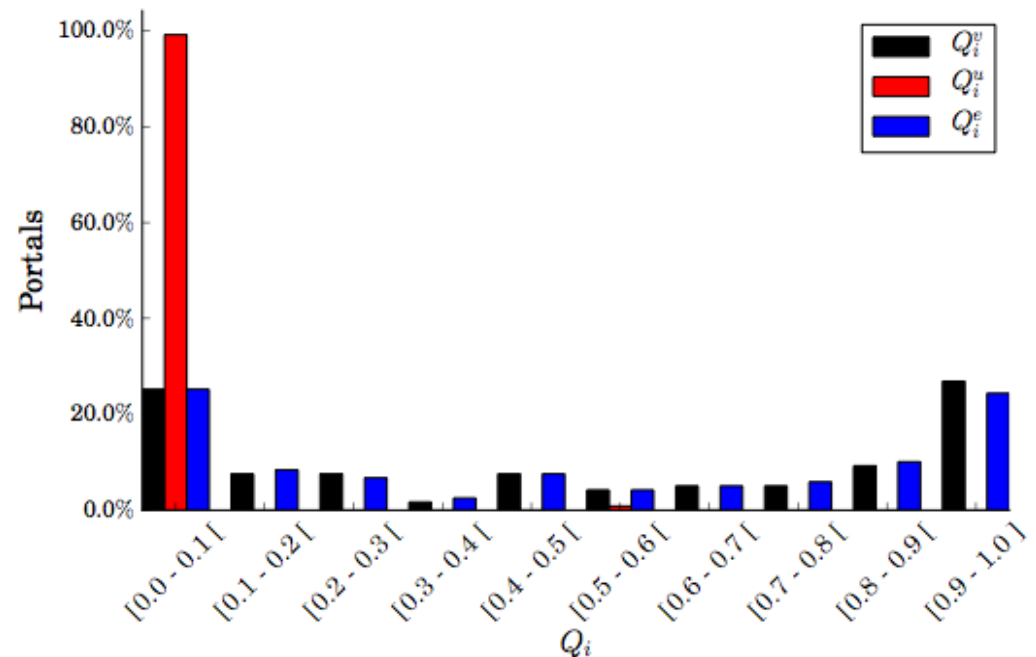
Contactability

Contact information in form of **URLs**, **email** addresses, or **any** value

very few URLs

35% of the portals with
very good contractibility

25% with hardly any
contact values



Conclusion

Main findings (126 CKAN Portals):

- High metadata **heterogeneity** for portal specific keys/tags
- Low confirmed openness (wrt. licenses and formats)
- About **80%** resource retrievability
- Only **35%** of the portals have a high contactability

Impact

Peer Reviewed Publications

- Jürgen Umbrich, Sebastian Neumaier, and Axel Polleres. Quality assessment & evolution of open data portals.
In *IEEE International Conference on Open and Big Data*, Rome, Italy, August 2015.
- Jürgen Umbrich, Sebastian Neumaier, and Axel Polleres. Towards assessing the quality evolution of open data portals.
In *ODQ2015: Open Data Quality: from Theory to Practice Workshop*, Munich, Germany, March 2015.

Follow-up Project: “**ADEQUATE**” [1]

- develop and evaluate mechanisms to **measure**, monitor and **improve** data quality in Open Data
- In cooperation with WU, Danube University Krems and Semantic Web Company

Current and Future Work

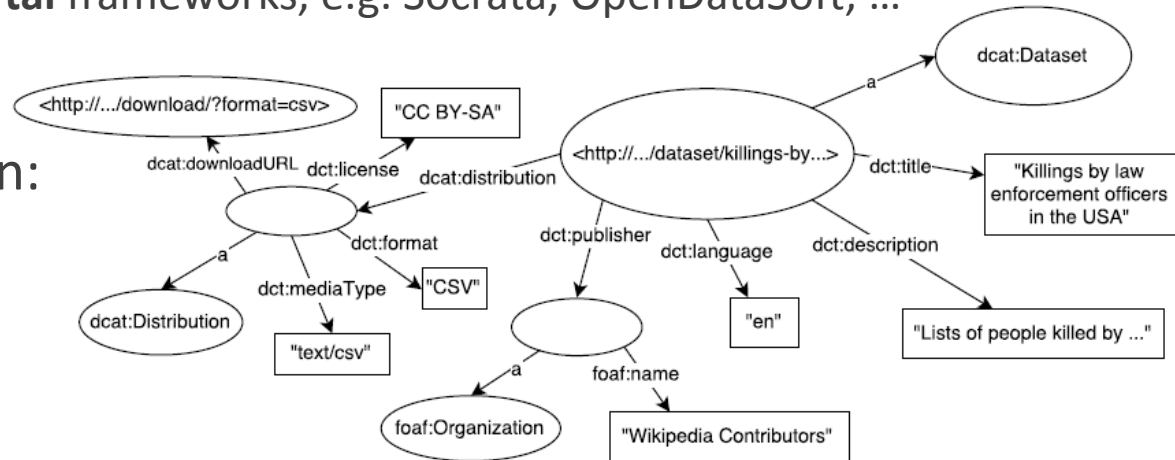
Towards a general QA Framework

More Open Data Portals:

- Harvest data from **other portal** frameworks, e.g. Socrata, OpenDataSoft, ...

Metadata Homogenization:

- Map metadata keys from different frameworks to the RDF-based **DCAT** [1]



DCAT specific Quality Dimensions:

- E.g., **Existence** and **conformance** of access, license or file format information.

Thank you for your attention.

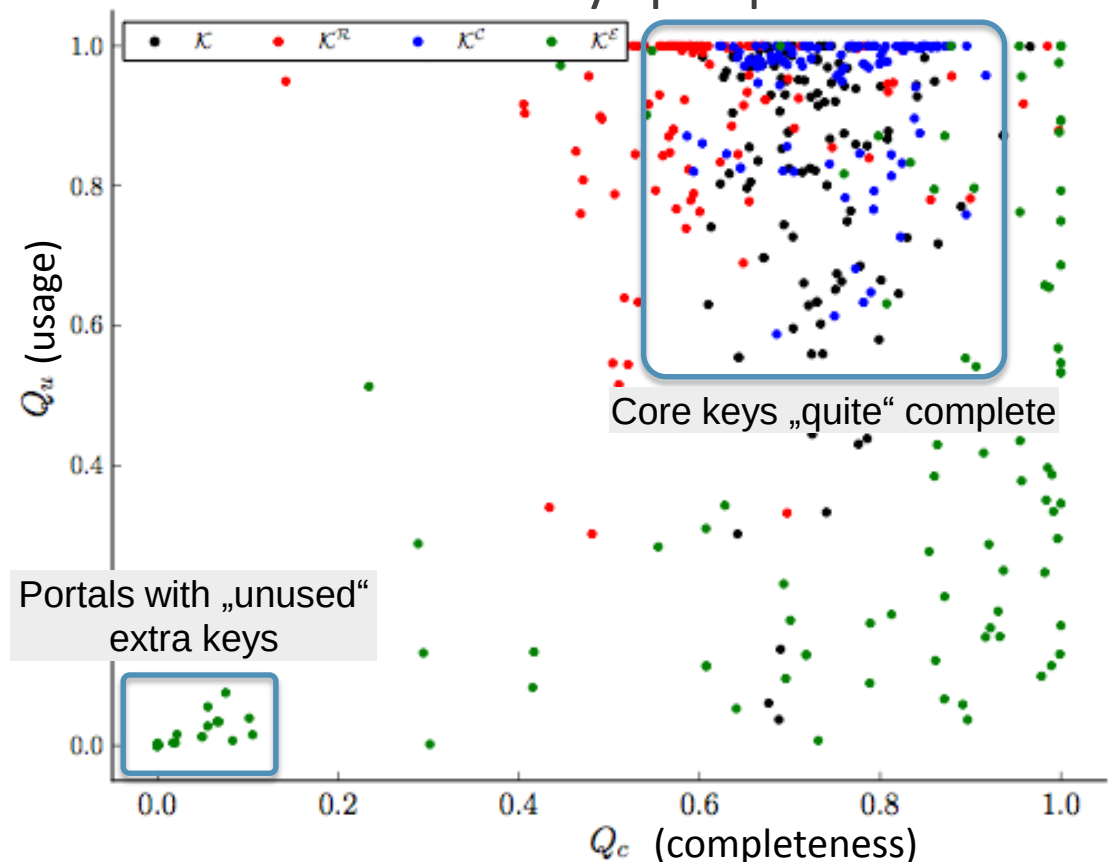
Backup Slides

Usage & Completeness

Avg. usage and completeness for different keys per portal

core and **resource**
keys are well
established

extra keys can be
grouped

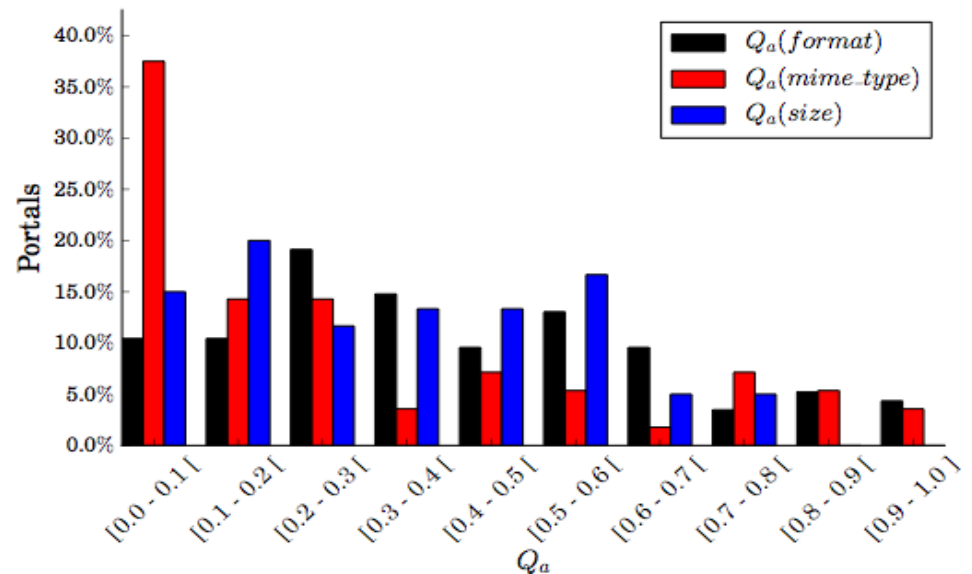


Accuracy

HTTP HEAD	1.64M	
response header	1.55M	94.5%
content-type	1.4M	85.4%
content-length	1.1M	67%

Datasets with metadata:

- 27K **size**
- 252K **mime type**
- 625K **format**



Formal Metrics (1/4)

Retrievability:

$$Q_r^{\mathcal{D}}(p) = \frac{\sum_{x \in \mathcal{D}(p)} \text{ret}(x)}{|\mathcal{D}(p)|}$$

$$Q_r^{\mathcal{R}}(p) = \frac{\sum_{x \in \mathcal{R}(p)} \text{ret}(x)}{|\mathcal{R}(p)|}$$

Usage:

$$\text{usage}(d) = \frac{|\text{keys}(d|\mathcal{K}^{\mathcal{C}} \cup \mathcal{K}^{\mathcal{E}})| + \sum_{r \in d} |\text{keys}(r)|}{|\text{keys}(p|\mathcal{K}^{\mathcal{C}} \cup \mathcal{K}^{\mathcal{E}})| + (|\text{keys}(p|\mathcal{K}^{\mathcal{R}})| * |\text{res}(d)|)}$$

$$Q_u(p) = \frac{\sum_{d \in \mathcal{D}(p)} \text{usage}(d)}{|\mathcal{D}(p)|}$$

Formal Metrics (2/4)

Completeness:

$$\text{compl}(k, d) = \begin{cases} 1 & \text{if } k \in \mathcal{K}^c \cup \mathcal{K}^e, \text{ and } \text{value}(k, d) \neq \text{Null} \\ \frac{\sum_{r \in \text{res}(d)} \text{compl}(k, r)}{|\text{res}(d)|} & \text{if } k \in \mathcal{K}^R \\ 0 & \text{otherwise} \end{cases}$$

$$\text{compl}(d) = \frac{\sum_{k \in \text{keys}(d)} \text{compl}(k, d)}{|\text{keys}(d)|}$$

$$Q_c(p) = \frac{\sum_{d \in \mathcal{D}(p)} \text{compl}(d)}{|\mathcal{D}(p)|}$$

Formal Metrics (3/4)

Accuracy:

$$\text{accr}(k, d) = \frac{\sum_{r \in \text{res}(d)} \text{accr}(k, r)}{|\text{res}(d)|}$$

$$Q_a(k, p) = \frac{\sum_{d \in \mathcal{D}(p)} \text{accr}(k, d)}{|\mathcal{D}(p)|}, \text{ with } k \in \mathcal{K}^*$$

Openness:

$$Q_o^l(p) = \frac{\sum_{d \in \mathcal{D}(p)} \text{open}_l(d)}{|\mathcal{D}(p)|}$$

$$Q_o^f(p) = \frac{\sum_{d \in \mathcal{D}(p)} \text{open}_f(d)}{|\mathcal{D}(p)|} \quad \{csv, html, latex, dvi, postscript, json, rdf, xml, txt, ical, rss, geojson, ods, ttf, otf, svg, gif, png\}$$

Formal Metrics (4/4)

Contactability:

$$Q_i(p) = \frac{\sum_{d \in \mathcal{D}(p)} \text{cont}(d)}{|\mathcal{D}(p)|}$$

$\text{cont}_e(d) = 1$ iff *value(maintainer_email, d) or value(author_email, d)*
is a valid email address (checked with regex)

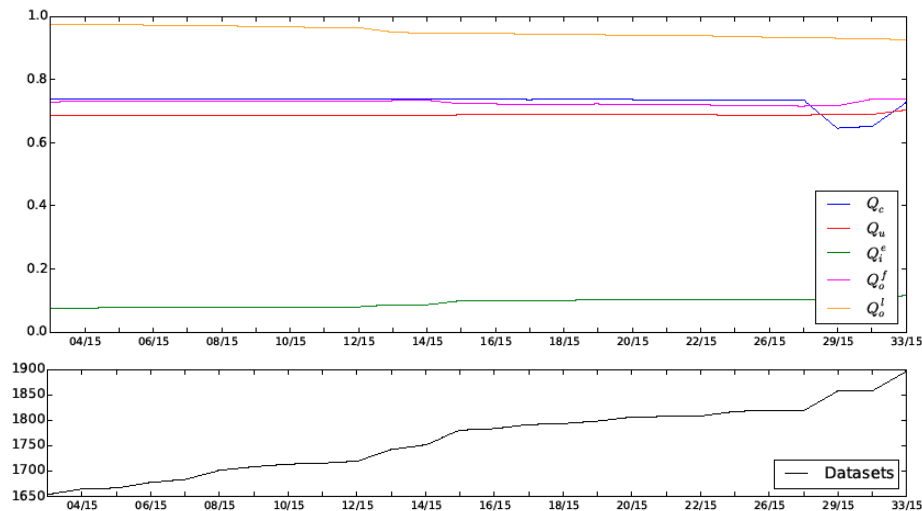
$\text{cont}_u(d) = 1$ iff *value(maintainer, d) or value(author, d)*
is a valid URL (checked with regex)

Portals Detail

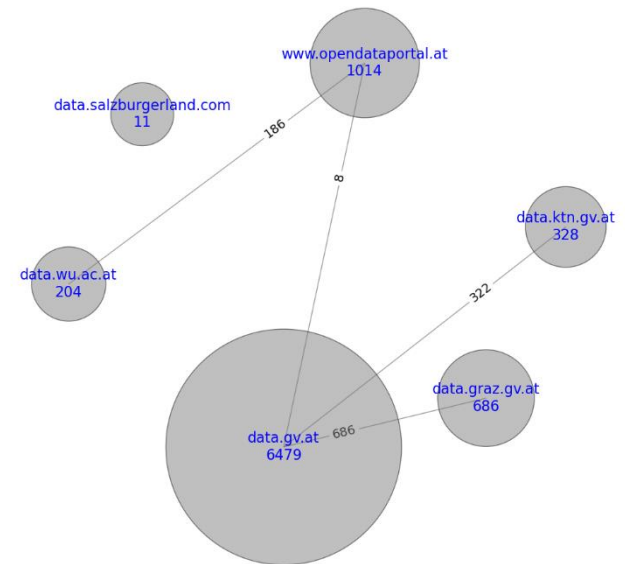
	$<10^2$	$<5 \times 10^2$	$<10^3$	$<10^4$	$<5 \times 10^4$	$<10^5$	$<10^6$	$>10^6$
$ \mathcal{D} $	35	42	10	32	2	3	2	0
$ \mathcal{R} $	18	31	19	38	13	3	3	1

Country	Total	CKAN	Socrata	OpenDataSoft
	239	126	102	11
US	112	20	90	2
UK	14	11	3	
Germany	9	9		
Italy	9	7	2	
France	9	2		7
Spain	8	7	1	
Austria	7	7		
Canada	7	5	2	
Australia	6	5	1	
Japan	5	5		
Brazil	5	5		
Ireland	4	3		1
Belgium	4	3		1
Netherlands	4	3	1	
Greece	3	3		
<i>others</i>	33	31	2	

Austrian Data Portals



Evolution of datasets and quality metrics



data.gv.at as harvesting portal